

FORD FULKERSON ALGORITHM

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You are given a directed flow network consisting of V vertices and E directed edges. Each edge has a capacity that represents the maximum amount of flow that can pass through it.

The source vertex is 0 and the sink vertex is $V-1$.

Your task is to find the maximum possible flow from the source to the sink using the Ford—Fulkerson algorithm.

Input Format

The first line contains two space-separated integers: V — number of vertices E — number of directed edges Each of the next E lines contains three space-separated integers: u — starting vertex v — ending vertex capacity — capacity of the edge

Constraints

$2 \leq V \leq 100$ $1 \leq E \leq 1000$ $0 \leq u, v < V$ $u \neq v$ $1 \leq \text{capacity} \leq 10^6$

Output Format

Print a single integer representing the maximum flow from vertex 0 to vertex $V-1$.

Sample Input 0

```
6 9
0 1 16
0 2 13
1 2 10
2 1 4
1 3 12
3 2 9
2 4 14
4 3 7
3 5 20
```

Sample Output 0

```
19
```

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Python 3



```
1 V, E = map(int, input().split())
2 graph = [[0] * V for _ in range(V)]
3 for _ in range(E):
4     u, v, c = map(int, input().split())
5     graph[u][v] += c
6 def dfs(u, sink, visited, flow):
7     if u == sink:
8         return flow
```

```

9         visited[u] = True
10        for v in range(V):
11            if not visited[v] and graph[u][v] > 0:
12                f = dfs(v, sink, visited, min(flow, graph[u][v]))
13                if f > 0:
14                    graph[u][v] -= f
15                    graph[v][u] += f
16                return f
17        return 0
18    source = 0
19    sink = V - 1
20    max_flow = 0
21    while True:
22        visited = [False] * V
23        flow = dfs(source, sink, visited, 10**9)
24        if flow == 0:
25            break
26        max_flow += flow
27    print(max_flow)
28

```

Line: 28 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

Download

```

1  6 9
2  0 1 16
3  0 2 13
4  1 2 10
5  2 1 4
6  1 3 12
7  3 2 9
8  2 4 14
9  4 3 7
10 3 5 20

```

Your Output (stdout)